Secure TCP Service Exposure via WireGuard VPN

Enterprise-Grade Networking Project

Author: Jacek
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Linux Networking • VPN • Firewall • NAT

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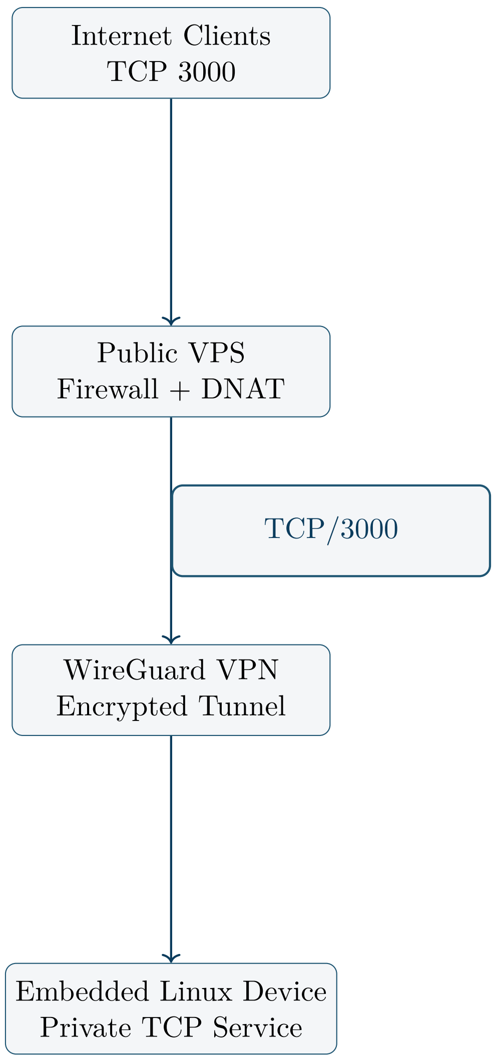
1 Project Overview

This document describes the design and implementation of an enterprise-grade network architecture used to securely expose a private TCP service running on an embedded Linux device via a public VPS. The embedded system is never directly exposed to the Internet; all traffic is routed through a controlled entry point and encrypted WireGuard tunnel.

2 Architecture Design

2.1 High-Level Diagram



2.2 Design Goals

- Zero direct Internet exposure of internal services
- Encrypted point-to-point communication
- Minimal attack surface on the public VPS
- Clear separation of public and private network layers

3 Technology Stack

- WireGuard VPN

- Linux networking (iptables, NAT, routing)
- VPS with public IPv4 address
- Embedded Linux (BusyBox-based)
- Diagnostic tools: tcpdump, netcat

4 Network Addressing

- VPS WireGuard address: 10.10.10.1
- Embedded device WireGuard address: 10.10.10.2
- Public TCP service port: 3000

5 Implementation

5.1 VPS Preparation

```
apt update
apt install -y wireguard iptables tcpdump netcat
sysctl -w net.ipv4.ip_forward=1
```

5.2 WireGuard Configuration (VPS)

```
[Interface]
Address = 10.10.10.1/24
ListenPort = 51821
PrivateKey = <VPS_PRIVATE_KEY>

[Peer]
PublicKey = <EMBEDDED_PUBLIC_KEY>
AllowedIPs = 10.10.10.2/32
```

5.3 Firewall and NAT Rules

```
iptables -I INPUT -p tcp --dport 3000 -j ACCEPT
iptables -I FORWARD -p tcp -d 10.10.10.2 --dport 3000 -j ACCEPT
iptables -t nat -A PREROUTING -p tcp --dport 3000 \
  -j DNAT --to-destination 10.10.10.2:3000
iptables -t nat -A POSTROUTING -o wg-oscam -j MASQUERADE
```

5.4 Service Requirements

The internal TCP service must listen on:

`0.0.0.0:3000`

6 Testing and Verification

```
nc -vz <PUBLIC_VPS_IP> 3000
tcpdump -i wg-oscam tcp port 3000
```

Successful tests confirm correct DNAT behavior and encrypted end-to-end TCP connectivity.

7 Result

The final solution delivers a stable, secure and production-ready architecture for exposing private TCP services through a public VPS while maintaining strict network isolation of the embedded device.

8 Skills Demonstrated

- Enterprise VPN architecture design
- Linux firewall and NAT configuration
- Packet-level TCP troubleshooting
- VPS administration
- Embedded Linux networking